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A sequence-based deep learning framework (PepInter) for protein–peptide interaction representation learning with pretrained protein language models

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Protein-peptide Interactions play essential roles in diverse cellular processes, yet their systematic identification remains challenging due to the limited availability of experimentally annotated protein-peptide interaction data. To address this challenge, we present PepInter, a sequence-based Deep Learning framework that leverages large-scale pretraining on structurally derived pseudo protein-peptide pairs to learn interaction-relevant representations. Specifically, energy-dominant peptide fragments are extracted from protein complexes curated from non-redundant Protein Data Bank structures, enabling the construction of pseudo protein-peptide interaction pairs that capture interface interaction patterns shared with canonical protein-protein interactions. This strategy allows the model to acquire interaction-aware priors in the absence of large-scale annotated protein-peptide complex datasets. Built upon the ESM Cambrian architecture, PepInter adopts a two-stage pretraining strategy. Following pretraining, PepInter is fine-tuned for both protein-peptide interaction classification and peptide bioactivity regression tasks. Across multiple benchmark datasets, PepInter consistently outperforms state-of-the-art methods and demonstrates strong generalization in identifying biologically meaningful PpIs. Case studies further highlight its ability to recover known interaction patterns and predict novel protein-peptide interactions. Together, these results establish PepInter as a scalable and effective framework for protein-peptide interaction prediction, with strong potential to accelerate peptide-based drug discovery.

The identification of potential interactions between peptides and proteins plays a crucial role in elucidating various biological functions, such as gene expression regulation and enzyme activity modulation[1]. Several studies have reported that peptides exhibit high specificity and relatively low immunogenicity, making them attractive candidates for therapeutic applications[2,3]. Moreover, the biological activity of peptides is mainly achieved through their interactions with proteins[4]. Some peptides can bind to protein or cell surface receptors selectively for activating or inhibiting signal transduction[5]. Therefore, predicting protein-peptide interactions (PpIs) is essential for understanding cellular processes and guiding the design of peptide-based therapeutics. Traditional experimental methods have been widely used to detect PpIs. However, they are often labor-intensive and time-consuming, requiring extensive procedures such as mutagenesis, binding assays and structural characterization, which limits their large-scale application[6]. Consequently, computational approaches for spredicting potential PpIs have emerged as an invaluable strategy to accelerate the discovery process[7–10]. These approaches can be broadly categorized into two classes: molecular docking-based and machine learning-based approaches. Docking-based methods[11–13] perform global searches over protein surfaces to identify potential peptide binding sites, followed by conformational sampling and scoring to select the optimal binding mode. Although these approaches can explore different conformational possibilities, their reliance on rigid scoring functions and limited sampling strategies constrains the ability to capture complex and nonlinear relationships between sequence and structure. Recent studies[14,15] show that, in blind prediction scenarios where the binding site is unknown or the interacting peptides exhibit high

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flexibility, docking-based predictions often struggle due to inefficiencies in sampling flexible interactions and the lack of learned sequence-level context. These limitations motivate the development of alternative computational approaches that can leverage large-scale sequence and structure information to infer interaction patterns beyond rigid conformational search.

Over the years, the growing availability of experimentally validated protein-peptide pairs has facilitated the development of traditional machine learning-based methods, such as SPRINT-seq[16] and PepBind[17]. However, their reliance on handcrafted features and limited generalization capabilities hinder their applicability to more complex and diverse prediction scenarios. Meanwhile, the rapid development of Deep Learning (DL) techniques[18,19] has greatly improved the accuracy of PpIs. For instance, Zhou et al. proposed a multi-task framework named PepGPL[20], which models local interactions between peptides and proteins. In this method, an interaction graph is constructed to capture interaction information from both the global sequence context and the local residue-level contexts of the sequence pairs. Sun et al. proposed a deep learning method called PepPAP, which is based on a convolutional neural network and multi-head attention. Sequence-based features such as sequence encoding, physicochemical properties, intrinsic disorder, and interface propensity have shown robust prediction performance[21]. With the rapid development of natural language processing (NLP) techniques, protein language models (PLMs) such as ESM[22], BERT[23] and ProtTrans[24] have been used to extract high-dimensional embeddings from amino acid sequences. Along this line, Jin et al. proposed a transformer-based model named TPepPro, which combines the features of local protein sequence with global structural information and employs an optimized BN-ReLU module that notably reduces the computational cost, showing great potential for PpI prediction[25]. Compared to docking-based methods, DL-based approaches are applicable to a wider range of proteins and peptides through informative sequence representations, which enables more efficient interaction screening and facilitates the discovery of new functional roles for both known and novel peptides.

A major obstacle in developing DL methods for modeling PpIs is the limited availability of high-quality training data. Experimentally validated PpIs datasets are typically small, sparsely distributed across protein families and biased toward a narrow set of well-studied targets, which poses several challenges for model training. Insufficient coverage of sequence and interaction space hinders the ability of models from learning generalizable interaction patterns, leading to overfitting and poor performance on unseen proteins or peptides. The lack of diverse interaction contexts limits the effectiveness of data-hungry architectures, particularly large sequence-based language models and graph-based neural networks, which rely on abundant and heterogeneous data to learn robust representations. As a result, models trained on limited experimental PpI data often fail to generalize beyond the training distribution and exhibit reduced predictive reliability. To mitigate these limitations, we draw inspiration from the "hot segment" paradigm of protein-protein interactions, in which interaction between two proteins is mediated by a short linear segment from one partner that contributes most of the interface energy[26]. Based on this principle, we construct a large-scale pseudo protein-peptide interaction pairs by leveraging known structures of protein-protein complexes, which serves as a natural and scalable data source for model pretraining. This systematically generated training dataset provides abundant supervision signals that help the model learn generalizable residue-residue interaction patterns. Pretrained interaction models can then be used to derive informative representations for protein-peptide sequence pairs, going beyond conventional single-sequence embeddings.

In this paper, we present PepInter, a sequence-based DL framework that integrates energy-guided structural priors with language model-based representation learning (Fig. 1). Specifically, we collect structure of protein-protein complexes from the public databases and apply the Rosetta Peptiderive protocol[27] to extract energy-dominant peptide fragments, thereby constructing pseudo protein-peptide pairs. These pairs are subsequently filtered to remove low-quality structures and fragments with negligible interface energy contributions, ensuring reliable interaction-relevant supervision. Energy-guided pretraining on structure-derived pseudo protein-peptide pairs endows PepInter with interaction-aware priors, which are further refined through ESMC-based (https://huggingface.co/EvolutionaryScale/esmc-600m-2024-12) masked language modeling (MLM) to integrate sequence semantics and latent interaction features. Subsequently, PepInter is fine-tuned on two complementary experimentally annotated datasets. The protein-peptide interaction dataset provides supervision for binary PpI classification, whereas the protein-peptide binding affinity dataset further refines the learned representations by modeling quantitative interaction strength, thereby ensuring that PepInter captures both interaction specificity and binding potency. Extensive evaluations using five-fold cross-validation and independent test sets demonstrate the superiority of PepInter, highlighting its potential as a powerful tool for large-scale PpI prediction.

Results

Comparative performance of embedding methods in protein-peptide interaction prediction

To investigate the impact of embedding strategies on interaction representation learning, we compare deep learning (DL) models using ESM2 and ESMC embeddings against machine learning (ML) models using ESMC embedding for PpI prediction. ESM2 variants (8 M, 35 M, 150 M, and 650 M parameters) and ESMC variants (300 M, 600 M) were evaluated. To ensure a fair comparison, all DL-based models employed identical downstream architectures (two fully connected layers) and training procedures with the same hyperparameters. For ML models, XGBoost[28] was selected as it outperformed other conventional methods in preliminary studies. As shown in Table 1, using PepInterDB with five-fold cross-validation, ESM2 models achieve F1 scores between 0.815 and 0.833, depending on the number of parameters, while ESMC models achieve 0.832, comparable to the best ESM2 model. As expected, XGBoost with EMSC embedding yields lower F1 scores of 0.798–0.803 and lowest recall metrics. These results confirm DL's advantage over ML, with larger parameter ESM2/ESMC models showing clear benefit. Given that ESM2_650M and ESMC_600M[29] show broadly comparable performance across evaluation metrics on PepInterDB, we selected ESMC_600M as the default encoder for PepInter, leveraging its newer model family and more compact parameterization, which are advantageous for large-scale inference and extensions.

Comparative performance with state-of-the-art methods in protein-peptide interaction prediction

To rigorously assess PepInter's effectiveness for PpI prediction, it is benchmarked against four state-of-the-art methods – MAARDTI[30], PIPR[31], HyperAttentionDTI[32], and Rep-ConvDTI[33] - as well as the ESM2_650M baseline model. For the ESM2_650M baseline, protein and peptide sequences were encoded using the pretrained ESM2_650M model with parameters frozen. The selected methods span a wide spectrum of modeling strategies, thereby ensuring that the comparison reflects both methodological diversity and current best practices. The details of these methods can be found in Supplementary Section S3. All methods were locally installed, trained, and evaluated under identical data splits on PepInterDB using five-fold cross-validation. The comparative performance of PepInter and the five competing methods is reported in Table 2, where the best-performing method for each metric is highlighted in bold.

Remarkably, PepInter achieves the best performance in the five-fold cross-validation experiments. It demonstrates stable improvements over the second-best model, ESM2_650M, across all performance measures. Specifically, PepInter shows slight improvement over ESM2_650M with ACC (1%), REC (2%), F1 (1%), and AUPR (1%), respectively. A higher recall indicates that the model is more effective in recognizing true interaction pairs. These results suggest that PepInter is capable of capturing semantic relationships between protein and peptide sequences, substantially enhancing the accuracy of PpI prediction.

Comparative performance in protein-peptide interaction prediction under different clustering settings

To assess the generalizability of PepInter to unseen cases, we repartitioned PepInterDB to reflect three scenarios: "Novel protein", "Novel peptide" and "Novel pair" based on similarity scores derived from Smith-Waterman alignment. As shown in Fig. 2, PepInter achieved consistently competitive performance across different clustering thresholds and evaluation settings. In particular, PepInter showed the best overall performance in the "Novel Protein" setting, with average Accuracy, AUC, and AUPR values of 0.89, 0.96, and 0.96 across thresholds from 0.3 to 0.6, respectively. PepInter also performed best in the more challenging "Novel Pair" setting, achieving average Accuracy, AUC, and AUPR values of 0.83, 0.92, and 0.90, respectively, outperforming the strongest baseline by 2.2%, 2.6%, and 3.9% in terms of Accuracy, AUC, and AUPR.

We further observed that the prediction performance of most methods decreased as the clustering threshold increased, suggesting that higher clustering thresholds introduced more challenging generalization scenarios. This trend was particularly evident in the "Novel Pair" setting, where the model was required to predict interactions involving both unseen protein clusters and unseen peptide clusters. Among the three settings, the "Novel Peptide" setting generally yielded higher performance for most methods, suggesting that peptide-cluster-based splitting may introduce a relatively smaller distribution shift in this benchmark. Nevertheless, PepInter consistently achieved the best overall performance in the "Novel Protein" and "Novel Pair" settings, and remained highly competitive in the "Novel Peptide" setting across different clustering thresholds. These results indicate that PepInter has strong and robust generalization ability under cluster-based evaluation. The detailed evaluation results are provided in Suppl. Section S5.

Evaluation of model robustness on ordered and disordered peptides

Considering the conformational flexibility of peptides, we further evaluated whether peptide disorder affects the predictive performance of PepInter. The model AIUPred[34] was used to estimate residue-level disorder propensity for peptide sequences in the PepInterDB dataset. Residues with an AIUPred disorder score (IDR) greater than 0.5 were considered disordered. For each peptide, if more than half of residues with disorder score greater than or equal to 0.5, was classified as disordered, and ordered otherwise[35,36]. For model robustness evaluation, the AIUPred-derived ordered/disordered labels were mapped to the held-out test samples in the five-fold cross-validation setting. As shown in Fig. 3A, B among the evaluation samples aggregated across the test dataset, 3432 peptide entries, accounting for 90.5%, were classified as disordered, while 362 peptide entries, accounting for 9.5%, were classified as ordered. This distribution indicates that the evaluated samples contain a large fraction of peptides with high predicted disorder propensity, highlighting the relevance of assessing model robustness under peptide conformational flexibility.

We stratified the original five-fold cross-validation predictions of PepInter and the baseline methods according to the AIUPred-derived ordered/disordered peptide labels. As shown in Fig. 3C–F and Table 3, PepInter maintained consistently strong performance on both peptide subsets. On the ordered peptide subset, PepInter achieved the best ACC, REC, and F1 among all methods, with values of 0.87, 0.75, and 0.83,

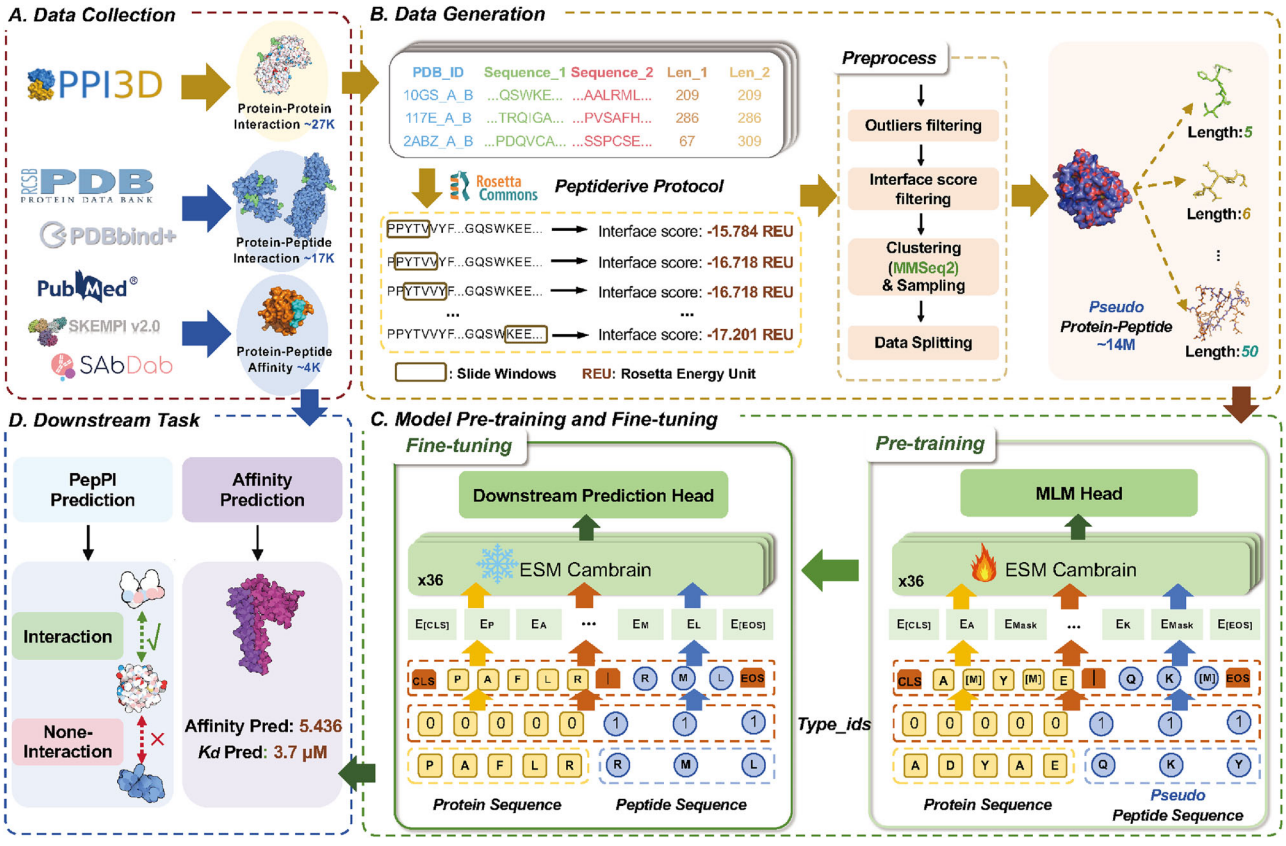


Fig. 1 | The workflow of PepInter. A Data collection: Protein-protein interaction data, protein-peptide affinity data, and protein-peptide interaction data are collected from several databases; (**B**) **Data generation:** Large-scale pseudo peptide-protein pairs are generated from protein complexes using the Rosetta Peptiderive protocol for subsequent pretraining; (**C**) **Model pre-training and fine-tuning:** The ESM-C model is adopted as the encoder backbone, and residue-level MLM is performed on the pseudo data pairs to learn implicit structural features within protein-peptide interaction space. For downstream tasks, PepPI and Affinity prediction heads are added and fine-tuned using experimentally validated protein-peptide data to adapt to specific prediction objectives; (**D**) **Downstream task:** PepInter enables both peptide-protein interaction prediction and binding affinity estimation.

Table 1 | Comparative performance of embedding methods in ML and DL models using five-fold cross-validation on PepInterDB

Encoding Methods	ACC	REC	F1	AUC	AUPR
$XGBoost_{ESMC_300M}$	0.81 ± 0.00	0.65 ± 0.01	0.80 ± 0.00	0.93 ± 0.00	0.94 ± 0.00
$XGBoost_{ESMC_600M}$	0.80 ± 0.01	0.64 ± 0.01	0.80 ± 0.01	0.93 ± 0.00	0.94 ± 0.00
ESM2_8M	0.82 ± 0.01	0.69 ± 0.01	0.82 ± 0.01	0.92 ± 0.00	0.93 ± 0.00
ESM2_35M	0.83 ± 0.00	0.71 ± 0.01	0.83 ± 0.00	0.93 ± 0.01	0.93 ± 0.01
ESM2_150M	0.83 ± 0.01	0.72 ± 0.01	0.83 ± 0.01	0.93 ± 0.01	0.94 ± 0.01
ESM2_650M	**0.84 ± 0.00**	**0.73 ± 0.01**	0.83 ± 0.01	**0.94 ± 0.01**	0.94 ± 0.01
ESMC_300M	0.83 ± 0.00	0.72 ± 0.01	0.83 ± 0.00	0.93 ± 0.01	0.94 ± 0.01
ESMC_600M	0.83 ± 0.00	0.71 ± 0.01	**0.83 ± 0.00**	0.93 ± 0.00	**0.94 ± 0.00**

All models are evaluated on the binary classification task of predicting whether a protein-peptide pair interacts. Results are reported as mean ± standard deviation across five folds. Higher values indicate better performance for all metrics.

Table 2 | Performance of PepInter in protein-peptide interaction prediction against the baseline and four state-of-the methods using five-fold cross-validation on PepInterDB

Models	ACC	REC	F1	AUC	AUPR
ESM2_650M	0.84 ± 0.00	0.73 ± 0.01	0.83 ± 0.01	0.94 ± 0.01	0.94 ± 0.01
MAARDTI	0.76 ± 0.01	0.55 ± 0.04	0.69 ± 0.26	0.87 ± 0.01	0.89 ± 0.01
PIPR	0.72 ± 0.01	0.52 ± 0.02	0.65 ± 0.02	0.81 ± 0.01	0.82 ± 0.01
Rep-ConvDTI	0.76 ± 0.01	0.57 ± 0.02	0.71 ± 0.01	0.85 ± 0.01	0.88 ± 0.00
HyperAttentionDTI	0.77 ± 0.01	0.57 ± 0.04	0.76 ± 0.02	0.88 ± 0.01	0.90 ± 0.01
PepInter	**0.85 ± 0.01**	**0.75 ± 0.01**	**0.84 ± 0.01**	**0.94 ± 0.01**	**0.95 ± 0.00**

Results are reported as mean ± standard deviation across five folds. The best result within each subset is highlighted in bold.

respectively. On the disordered peptide subset, PepInter also achieved the best ACC, REC, and F1, with values of 0.85, 0.75, and 0.87, respectively. These results suggest that PepInter preserves stable classification performance across both ordered and disordered peptide subsets. The ROC and PR curves further support this observation. PepInter achieved an AUC of 0.938 and an AUPR of 0.946 on the disordered peptide subset, outperforming the compared methods. On the ordered peptide subset, PepInter achieved an AUC of 0.932 and an AUPR of 0.935. Although ESM2_650M showed a slightly higher AUC and comparable AUPR on ordered peptides, PepInter achieved superior threshold-dependent metrics, including ACC, REC, and F1. Overall, these results indicate that PepInter maintains stable predictive performance across both ordered and disordered peptide subsets, suggesting reasonably robustness to peptide conformational flexibility in the evaluated PepInterDB samples.

Evaluation of model generalizability on imbalanced datasets

To evaluate the robustness of PepInter in binary interaction prediction and to simulate the scarcity of positive samples commonly observed in real biological experiments, we constructed 4 imbalanced data distributions in the training set while keeping the original distribution unchanged in the test set. This ensures that the evaluation results faithfully reflect the model's performance in realistic application scenarios. Under different positive/negative sample ratios, we compared the prediction performance of PepInter with the ESM2_650M and state-of-the-art methods on these additional training sets. Specifically, we randomly paired peptides and proteins in the training set to generate negative samples for which no known interaction evidence exists. Four imbalanced datasets with ratios of 1:4, 1:9, 1:19, and 1:49 were generated. All models were trained and evaluated under identical experimental settings for fair comparison.

As shown in Fig. 4A, B, PepInter achieves the best performance across all imbalanced test scenarios, demonstrating superior robustness compared with other models. As the positive-to-negative ratio decreases from 1:1 to 1:49, all models exhibit declining accuracy, underscoring the challenges posed by severe class imbalance (Fig. 4A). Nevertheless, PepInter maintains the highest accuracy and accuracy across all imbalance settings. As shown in Fig. 4B, PepInter consistently achieves the highest F1 score across all positive-to-negative ratios and maintains stronger performance even under increasingly imbalanced training conditions. However, under extreme imbalance conditions, models such as PIPR and Rep-ConvDTI suffer from substantial performance degradation. In contrast, PepInter demonstrates consistently strong performance even when positive samples are highly sparse, highlighting its superior robustness and generalization ability.

To further investigate whether PepInter's attention mechanism can implicitly reflect structural adaptability, we selected two PDB complexes, 3EJH and 9DPF, as illustrative examples and analyzed the relationship between normalized cross-attention weights and crystallographic B-factors, which were used as an experimental proxy for local residue flexibility. As summarized in Fig. 4C, the attention weights showed positive correlations with B-factor-derived flexibility in both complexes. Specifically, Spearman correlation analysis yielded $r = 0.454$ with $p = 6.03 \times 10^{-6}$ for 3EJH and $r = 0.216$ with $p = 9.49 \times 10^{-4}$ for 9DPF. In addition, residues classified as flexible exhibited higher mean attention values than rigid residues in both structures, and the differences were statistically significant according to the Mann-Whitney U test. For 3EJH, Fig. 4D, F, G, H show the per-residue line plot, attention/B-factor heatmap, scatter plot, and rigid-versus-flexible distribution comparison, respectively. For 9DPF, the corresponding analyses are shown in Fig. 4E, I, J, K. The line plots and heatmaps indicate spatial agreement between regions with elevated B-factors and higher cross-attention weights along the receptor sequences. The scatter plots further show positive associations between normalized B-factors and attention weights at the residue level, whereas the violin/box plots demonstrate that flexible residues generally receive higher attention than rigid residues. To complement the flexibility analysis, we further examined whether the attention weights also reflect structural proximity. Pairwise $C_\alpha - C_\alpha$

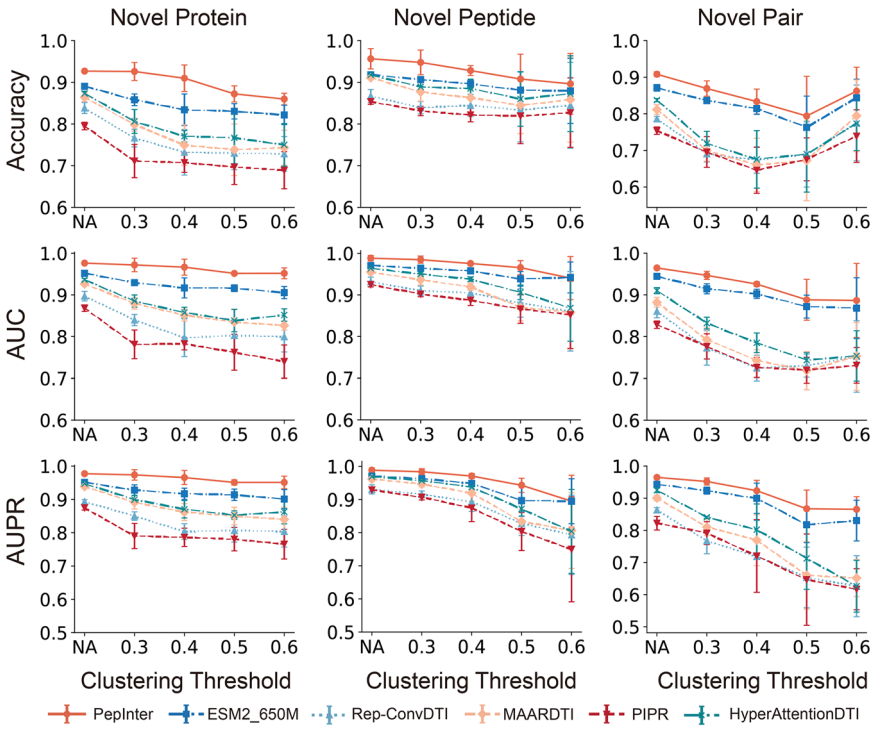


Fig. 2 | Performance comparison of PepInter and baseline methods under different cluster-based evaluation scenarios. Models were evaluated under the "Novel Pair", "Novel Protein", and "Novel Peptide" settings across sequence similarity thresholds of NA, 0.3, 0.4, 0.5, and 0.6. Here, "NA" denotes random cross-validation, in which 80% of the dataset was randomly used for training, and the remaining 20% was used for performance evaluation. Error bars in the "Novel Protein", "Novel Peptide" and "Novel Pair" settings indicate the mean ± standard deviation over five folds ($n = 5$).

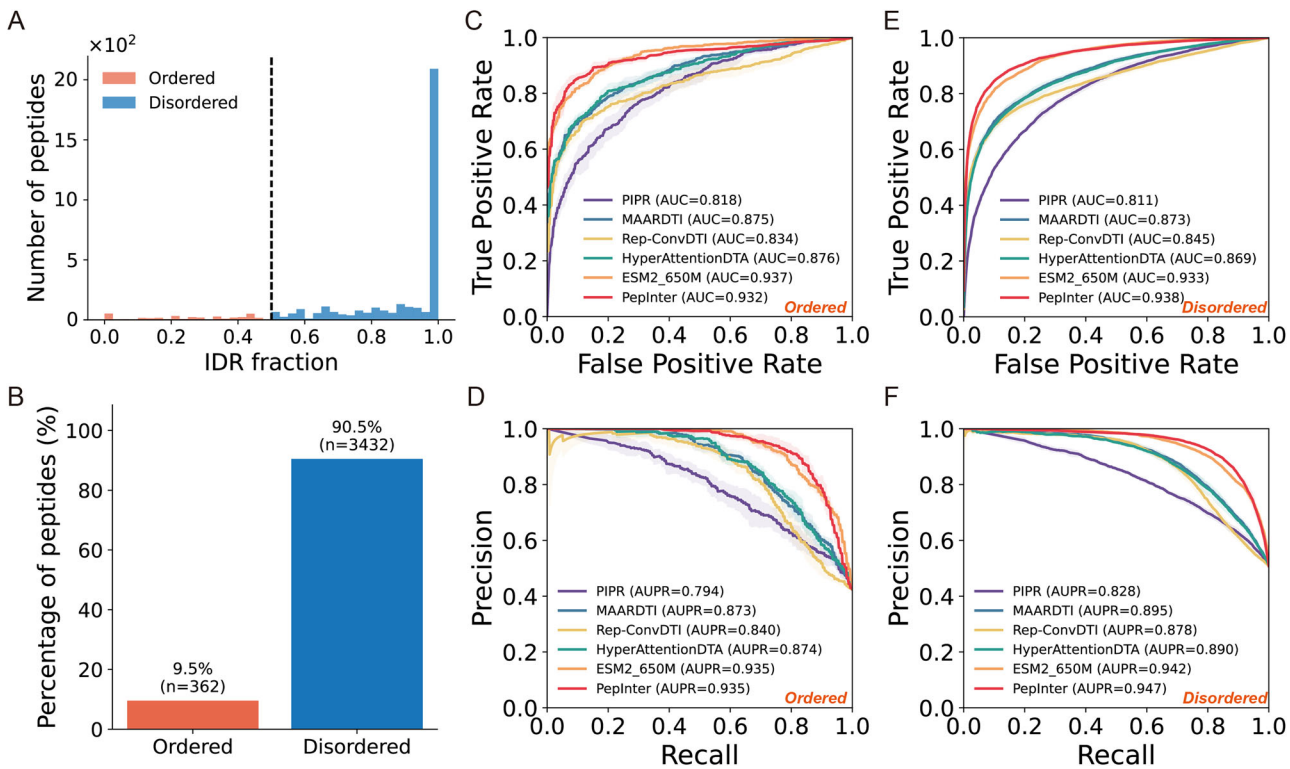


Fig. 3 | AIUPred-based analysis of peptide disorder and stratified model performance on PepInterDB. A Distribution of IDR fractions for peptide entries in the evaluation set. Residues with AIUPred disorder scores greater than 0.5 were considered disordered, and the IDR fraction of each peptide was calculated as the proportion of predicted disordered residues. The dashed line indicates the IDR fraction threshold of 0.5 used to classify peptides as ordered or disordered. **B** Proportion of ordered and disordered peptide entries in the evaluation set. **C**, **E** ROC curves for the compared models on ordered and disordered peptide subsets, respectively. Curves represent the mean performance across $n = 5$ independent experiments based on five-fold cross-validation, and shaded areas indicate the standard deviation. **D**, **F** AUPR curves for the compared models on ordered and disordered peptide subsets, respectively. Curves represent the mean performance across $n = 5$ independent experiments based on five-fold cross-validation, and shaded areas indicate the standard deviation.

Table 3 | Stratified classification performance of PepInter and baseline methods on ordered and disordered peptide subsets of PepInterDB using five-fold cross-validation

Type	Models	ACC	REC	F1
Ordered	ESM2_650M	0.85 ± 0.01	0.69 ± 0.02	0.80 ± 0.02
	MAARDTI	0.79 ± 0.02	0.54 ± 0.04	0.68 ± 0.03
	PIPR	0.75 ± 0.01	0.51 ± 0.01	0.63 ± 0.01
	Rep-ConvDTI	0.79 ± 0.01	0.58 ± 0.04	0.71 ± 0.03
	HyperAttentionDTI	0.80 ± 0.01	0.56 ± 0.03	0.70 ± 0.02
	PepInter	**0.87 ± 0.01**	**0.75 ± 0.02**	**0.83 ± 0.02**
Disordered	ESM2_650M	0.82 ± 0.01	0.67 ± 0.01	0.79 ± 0.01
	MAARDTI	0.75 ± 0.02	0.55 ± 0.04	0.69 ± 0.03
	PIPR	0.71 ± 0.00	0.52 ± 0.01	0.65 ± 0.01
	Rep-ConvDTI	0.77 ± 0.00	0.58 ± 0.01	0.72 ± 0.01
	HyperAttentionDTI	0.75 ± 0.01	0.54 ± 0.02	0.68 ± 0.01
	PepInter	**0.85 ± 0.01**	**0.75 ± 0.02**	**0.84 ± 0.01**

Results are reported as mean ± standard deviation across five folds. Ordered and disordered peptide subsets were defined according to AIUPred-predicted IDR fractions. The best result within each subset is highlighted in bold.

distances between all peptide-receptor residue pairs were computed and Spearman correlations with the corresponding cross-attention weights were calculated. For 3EJH, a moderate negative correlation was observed $\rho = -0.403, p < 10^{-70}$, indicating that higher attention tends to be directed toward spatially proximal residue pairs. For 9DPF, the correlation was weak and in the opposite direction $\rho = +0.106, p < 10^{-11}$. Taken together, the B-factor and distance analyses suggest that PepInter's cross-attention mechanism captures a combination of structural flexibility and spatial proximity, with the relative contribution varying across complexes. In addition, Fig. 4L, M further compare residue-level attention patterns with representative protein-peptide interface regions for 3EJH and 9DPF, respectively. The attention maps highlight several high-attention regions located near the experimentally observed peptide-binding interfaces, and the corresponding local structural views show representative receptor and peptide residues involved in these interface regions.

Together, these results indicate that PepInter maintains robust predictive performance under severe class imbalance and that its cross-attention mechanism can preferentially emphasize structurally flexible regions. Although PepInter does not explicitly perform conformational sampling or dynamic simulations, this attention pattern suggests that the model can partially capture flexibility-associated interaction signals from static structural inputs, thereby providing a plausible explanation for its robustness and generalization ability in challenging protein-peptide interaction prediction scenarios.

Comparative performance in protein-peptide affinity prediction using PepAffDB

Protein-peptide binding affinity is a key determinant of biological activity, interaction specificity, and downstream functional outcomes. Accurate prediction of protein-peptide affinity holds significant scientific importance and practical values in biomedical research and therapeutic development. Here, we conducted a systematic evaluation of predictive approaches using a curated protein-peptide affinity dataset to assess model performance. Similar to the binary interaction prediction setting, the state-of-the methods (DeepDTA[37], TEFDTA[38], AttentionDTA[39]) and the ESM2_650M baseline were locally installed, trained, and tested using five-fold cross-validation on the same dataset to ensure a fair and unbiased performance comparison. As shown in Table 4, PepInter exhibits the lowest prediction errors among all compared methods, with an RMSE of 1.02 and an MAE of 0.77, representing a 10% reduction in prediction error compared to the second-best model, ESM2_650M. Furthermore, it achieves the highest R^2 value of 0.65, representing a 9% increase over ESM2_650M in its ability to explain the variance in the binding affinity data.

The scatter plots of the compared models in Fig. 5 provide further visual evidence for the superiority of PepInter. Specifically, the predictions generated by PepInter are distributed more tightly around the diagonal reference line, indicating a closer agreement between predicted and true affinity values. By contrast, ESM2_650M and DeepDTA show noticeably broader scatter distributions, with more points deviating from the diagonal line, reflecting weaker fitting performance and larger prediction uncertainty. TEFDTA and AttentionDTA demonstrate moderate improvements, but their prediction distributions remain less compact than that of PepInter. Furthermore, the marginal histograms in each panel reveal that the prediction distribution produced by PepInter is more closely aligned with the distribution of the true values, suggesting improved calibration and reduced systematic bias. Overall, both the quantitative metrics and the qualitative visualization consistently indicate that PepInter provides the most accurate, robust, and reliable affinity prediction performance among the compared methods, highlighting its effectiveness in modeling PpIs.

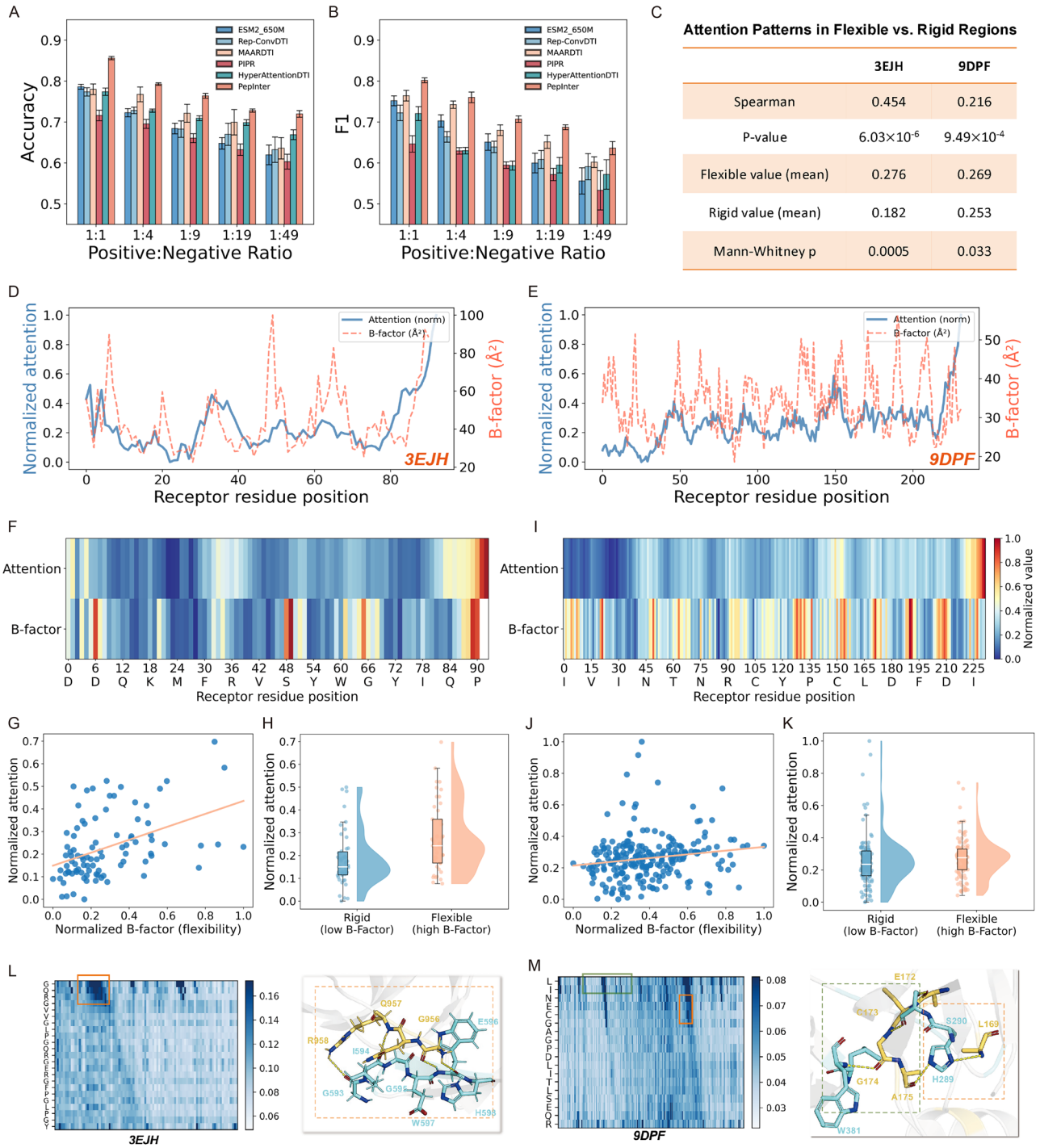
Given that PepAffDB contains affinity records reported as *Kd*, *Ki*, and *IC50*, we further examined whether the unified affinity formulation introduced measurement-type-specific performance differences. Although these values are commonly converted to a pAffinity scale for model training, they represent related but non-identical biochemical measurements and may differ in their dependence on experimental conditions. We therefore stratified the held-out predictions of PepInter according to the original measurement type and recalculated the regression metrics for each subset.

The evaluation set was dominated by *Kd* measurements, with 226 *Kd* records, followed by 66 *IC50* records and only 9 *Ki* records. PepInter achieved an RMSE of 1.05, an MAE of 0.79, an R^2 of 0.55, and a Spearman correlation of 0.70 on the *Kd* subset. For the IC50 subset, PepInter showed lower prediction errors and higher correlation, with an RMSE of 0.77, an MAE of 0.63, an R^2 of 0.84, and a Spearman correlation of 0.92. In contrast, the *Ki* subset showed higher prediction errors, with an RMSE of 1.49 and an MAE of 1.27, although the Spearman correlation remained 0.79. Because the *Ki* subset contained only nine samples, these values should be interpreted cautiously.

Overall, the measurement-type stratified analysis indicates that PepInter retains predictive capacity across different affinity measurement types, particularly for the two major subsets, *Kd* and *IC50*. At the same time, the variation among measurement types suggests that combining *Kd*, *Ki*, and *IC50* into a single affinity prediction task may introduce a degree of label heterogeneity. This limitation is especially relevant for underrepresented measurement types such as *Ki*. Future extensions of PepAffDB-based affinity modeling may benefit from explicitly incorporating measurement type or assay metadata as auxiliary information.

Case Study 1 - Semaglutide and GLP-1 receptor identification

The glucagon-like peptide-1 receptor (GLP-1R) is one of the most effective therapeutic targets for the treatment of type 2 diabetes mellitus[40]. It can effectively activate multiple downstream signaling pathways, including PKA, PI3K, and MAPK, and is involved in important physiological processes such as insulin secretion and β-cell proliferation[41]. Therefore, GLP-1R, as the functional target through which GLP-1 exerts its effects, has potential significance for guiding the study of multiple diseases. In this case study, PepInter was evaluated for its ability to identify the protein target GLP-1R of semaglutide and seven of its analogs among the proteins in the PepInterDB dataset[42]. To prevent data leakage, PepInter was retrained on a filtered dataset excluding semaglutide-like peptides and GLP-1R-like protein sequences. Specifically, protein sequences exceeding 40% similarity to these targets were removed from the training set, yielding a cleaned dataset of 13,187 protein-peptide pairs for model training. The test set comprised all pairwise combinations of 2940 proteins and the seven semaglutide analog peptides, generating 20,580 candidate protein-peptide pairs. The known



pairs, GLP-1R and semaglutide-analog peptides, were assigned binary labels of 1, while the remaining candidate pairs were labeled 0.

Table 5 presents the predicted probabilities for semaglutide-analog peptide-GLP-1R pairs by PepInter. All positive pairs receive high predicted probabilities ranging from 0.984 to 1.000, and GLP-1R is consistently ranked within the top 5% of candidate proteins for all semaglutide analogs. Since this target-identification task is highly imbalanced, with only 7 positive pairs among 20,580 candidate pairs, we further reported AUCPR in addition to AUC. PepInter achieved an AUC of 0.982 and an AUCPR of 0.0356, whereas the random AUCPR baseline is approximately 0.00034. The much lower AUCPR compared with AUC confirms that AUC alone can be optimistic under severe class imbalance. Nevertheless, the AUCPR is approximately 105-fold higher than the random baseline, supporting meaningful enrichment of known GLP-1R interactions among top-ranked candidates. To further characterize the embedding representations generated by PepInter, UMAP was applied to project candidate protein-peptide pairs into a two-dimensional space (Suppl. Fig. S2).

Case study 2 - Parathyroid hormone and PTH type 1 receptor identification

Parathyroid hormone receptor type 1 (PTH1R) is a member of the G protein-coupled receptor family and is widely expressed in vivo, where it participates in multiple biological processes, including systemic metabolism and tumorigenesis[43]. Parathyroid hormone (PTH) and parathyroid hormone-related peptide (PTHrP) are key regulators of calcium and phosphate metabolism in the body. They also play important roles in the

Fig. 4 | Prediction performance, attention weight distributions, and the relationship between cross-attention and structural flexibility. A, B Show the performance comparison of PepInter and baseline methods in protein-peptide interaction prediction under varying positive/negative ratios, evaluated by Accuracy and F1 score, respectively. The bars represent the mean performance across $n = 5$ independent experiments based on five-fold cross-validation, and error bars indicate the standard deviation. **C** Summarizes attention weight statistics relative to B-factor-derived residue flexibility for representative complexes 3EJH and 9DPF. Spearman correlation coefficients and associated p-values from a two-sided Spearman correlation test quantify the correlation between residue flexibility and attention weights; mean attention values for residues classified as flexible versus rigid are reported. **D**, **E** Plot per-residue normalized cross-attention weights (blue line) and crystallographic B-factors (red dashed line) along the receptor sequence for 3EJH ($n = 91$ independent residues) and 9DPF ($n = 232$ independent residues), respectively. Corresponding heatmaps of normalized attention and normalized B-factors across receptor residue positions are shown in (**F**) for 3EJH and (**I**) for 9DPF, with color scales ranging from low (blue) to high (red) and residue identity annotated at selected positions. Scatter plots in (**G**) for 3EJH and (**J**) for 9DPF display the relationship between normalized attention and normalized B-factor for individual residues, with red lines indicating linear regression fits; positive Spearman correlations (3EJH, $r = 0.454$, $p = 6.03 \times 10^{-6}$; 9DPF, $r = 0.216$, $p = 9.49 \times 10^{-4}$) suggest a tendency for higher attention in more flexible regions. (**H**) for 3EJH and (**K**) for 9DPF compare attention distributions between rigid low-B-factor residues and flexible high-B-factor residues. One-sided Mann–Whitney U tests (flexible > rigid) were used to assess whether flexible residues receive significantly higher attention weights than rigid residues (3EJH, $p = 5.0 \times 10^{-4}$; 9DPF, $p = 0.033$). The plots include $n = 91$ independent residues for 3EJH and $n = 232$ independent residues for 9DPF, respectively. (**L**, **M**) Further visualize the residue-level attention patterns and representative interface regions for 3EJH and 9DPF, respectively. The left subpanels show the attention between receptor residues and peptide residues, where darker colors indicate higher attention weights. The right subpanels show the corresponding local protein-peptide interface structures, with representative interacting residues highlighted.

Table 4 | Performance comparison of PepInter in protein-peptide binding affinity prediction with four the state-of-the-art methods using five-fold cross-validation on the PepAffDB dataset

Models	RMSE↓	MAE↓	R^2↑	Spearman↑
DeepDTA	1.29 ± 0.06	0.98 ± 0.07	0.44 ± 0.05	0.67 ± 0.03
TEFDTA	1.22 ± 0.06	0.91 ± 0.04	0.47 ± 0.05	0.71 ± 0.02
AttentionDTA	1.21 ± 0.07	0.90 ± 0.08	0.51 ± 0.06	0.71 ± 0.02
ESM2_650M	1.14 ± 0.03	0.87 ± 0.03	0.56 ± 0.02	0.72 ± 0.01
PepInter	**1.02 ± 0.03**	**0.77 ± 0.03**	**0.65 ± 0.02**	**0.79 ± 0.01**

treatment of osteoporosis by increasing bone mass, promoting bone formation, and reducing the incidence of high-risk fractures. Five PTH and PTHrP analogs are known to date to bind to PTH1R[18]. The same procedure described above was applied to remove protein and peptide sequences with sequence similarity greater than 40%. Based on this filtering strategy, the remaining 2922 proteins were paired with the PTH and PTHrP analogs to construct an independent test set, comprising a total of 14,615 protein-peptide pairs. These pairs were subjected to PepInter prediction and ranked according to their predicted probabilities. Table 6 shows the predicted interaction probabilities of the five known PTH and PTHrP peptides with PTH1R, which range from 0.887 to 1.000. PTH1R was consistently ranked within the top 5% of candidate proteins for all PTH analog peptides. To account for severe class imbalance, we additionally reported AUCPR. PepInter achieved an AUC of 0.9662 and an AUCPR of 0.0180, approximately 53-fold higher than the random AUCPR baseline, supporting enrichment of known PTH1R-related interactions among top-ranked candidates.

AlphaFold-Multimer-based structural validation and HPEPDOCK case-level comparison

To further assess the structural plausibility of the case-study predictions and to clarify the complementary roles of PepInter and structure-based prediction methods, we performed structural validation using AlphaFold-Multimer[44] for two representative systems, Semaglutide–GLP-1R and PTH/PTHrP–PTH1R. Because AlphaFold-Multimer requires full complex structure prediction and is computationally intensive, it is not practical for exhaustive large-scale screening of candidate pairs. Therefore, it was applied here only as a downstream validation tool for selected representative cases.

For each case, negative pairs were selected using a stratified random sampling strategy. Specifically, all non-homologous protein–peptide pairs were grouped by their PepInter-predicted probabilities into five equal-width intervals: [0, 0.2], [0.2, 0.4], [0.4, 0.6], [0.6, 0.8], and [0.8, 1.0]. Up to 30 samples were randomly drawn from each interval, resulting in approximately 150 background negative pairs. This approach ensured that the negative set covered the full range of PepInter prediction scores. In addition, for each peptide, one representative stringent negative pair was selected from the original validation set for highlighted comparison. These stringent negative pairs corresponded to non-homologous receptor pairs assigned high probabilities by PepInter. This way enabled direct comparison between known positive pairs and representative high-scoring negative pairs, while the background negative pairs were to estimate the overall ipTM score distribution.

As shown in Fig. 6, AlphaFold-Multimer ipTM scores exhibited clear distributional differences between known positive pairs and background negative pairs. In the Semaglutide–GLP-1R system, all seven known positive pairs achieved high ipTM scores, ranging from 0.84 to 0.90. By contrast, the 150 background negative pairs showed a median ipTM score of 0.36, with 83.3% of them (125/150) below 0.5. Notably, although the highlighted stringent negative pairs all received high PepInter probabilities greater than 0.89, their ipTM scores ranged from 0.61 to 0.85 and were generally lower than those of the corresponding positive pairs. These results suggest that AlphaFold-Multimer can provide additional structural-level confidence for high-scoring candidates that are difficult to distinguish using sequence-based prediction alone.

A similar trend was observed in the PTH/PTHrP–PTH1R system. The five known positive pairs yielded consistently high ipTM scores within the range of 0.88–0.91, whereas the background negative pairs had a median ipTM score of 0.38, with 77.3% of them (116/150) below 0.5. Among the highlighted stringent negative pairs, four showed ipTM scores below 0.5, ranging from 0.40 to 0.49. The only exception was the negative pair corresponding to *Neg_7*, which achieved an ipTM score of 0.79, indicating potential structural ambiguity for this specific pairing. In terms of computational efficiency, PepInter can complete 10 predictions in 1 s, whereas AF2-Multimer requires 1 h for the same number of cases, corresponding to an approximately 3600-fold improvement in speed.

Overall, the distributional separation in ipTM scores between known positive pairs and background negative pairs across the two representative systems supports the utility of structure-based modeling as a downstream validation step following sequence-based screening. These results indicate that PepInter is suitable for rapid large-scale prioritization of protein–peptide candidate pairs, whereas AlphaFold-Multimer can provide additional structural-level evidence for the prioritized candidates. Therefore, the two approaches are best utilized as complementary components in a hierarchical screening workflow rather than as competing prediction strategies.

In addition to AlphaFold-Multimer, structure-based protein-peptide docking provides a baseline for comparison. Here, Hpepdock[11] was selected as a representative protein–peptide docking tool, which supports both blind and known binding-site docking scenarios, without requiring case-specific restraint definition, in contrast to information-driven approaches such as

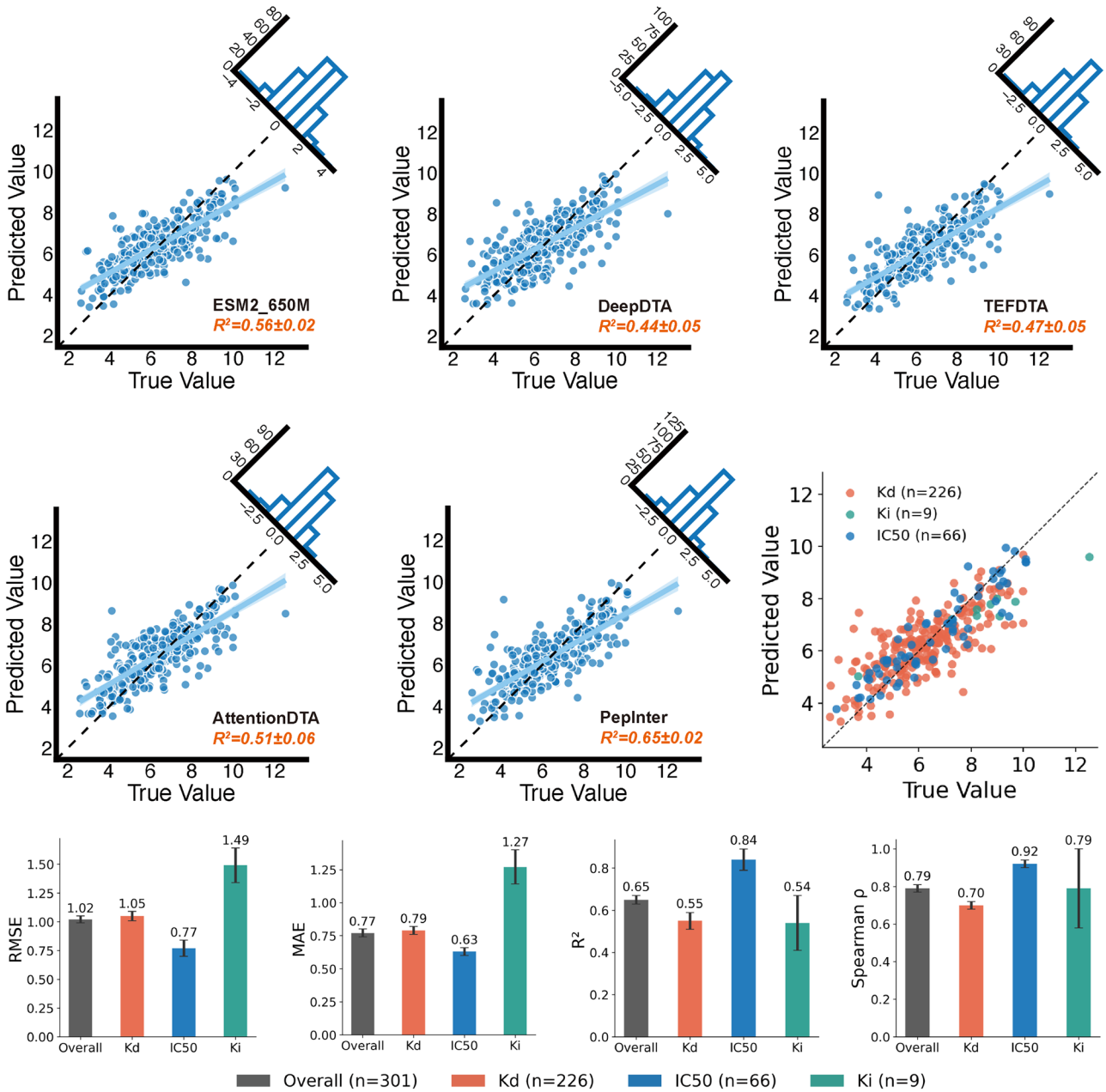


Fig. 5 | Performance comparison and measurement-type stratified evaluation on PepAffDB. Predicted versus true binding affinity values are shown for ESM2_650M, DeepDTA, TEFDTA, AttentionDTA, and PepInter. The dashed diagonal line indicates perfect prediction, and the marginal histograms show the distributions of true and predicted affinity values. The lower panels summarize the measurement-type stratified performance of PepInter across Overall, *Kd*, *IC50*, and *Ki* subsets, including RMSE, MAE, R^2, and Spearman correlation. The number of samples in each subset is indicated in the legend. Bars represent the mean performance across five-fold cross-validation, and error bars indicate the standard deviation.

Table 5 | Predicted interaction probabilities between semaglutide analogs and GLP-1R

PDB	Peptide Sequence	Probability
3IOL	HAEGTFTSDVSSYLEGQAAKEFIAWLVKGRG	0.988
4APD	HAEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	0.984
5OTT	HXEGXFTSDLSKQMEEEAVRLFIEWLKNGGPSSGAPPPS	0.992
5OTU	HXEGXFTSDVSSYLEGQAAKEFIAWLVKGRG	0.996
5OTV	HCEGXFTSDVSSYLEGQAAKEFIAWLVKGRG	0.996
5OTW	HXEGCFTSDVSSYLEGQAAKEFIAWLVKGRG	1.000
5OTX	HCEGCFTSDVSSYLEGQAAKEFIAWLVKGRG	1.000

Table 6 | Predicted interaction probabilities between PTH analogs and PTH1R

PDB	Peptide Sequence	Probability
1BWX	SVSEIQLMHNLGKHLNSMERVEWLRKKLQDVHNF	0.996
6FJ3	XVAEIQLMHQRAKWLNSLERVEWLRKKLQDVHNYX	1.000
6NBF	AVAEIQLMHQRAKWIQDARRRAFLHKLIAEIHTAEI	1.000
7VVJ	AVSEHQLLHDKGKSIQDLRRRFFLHHLIAEIHTAEI	0.887
8FLS	AVSEHQLLHDKGKSIQDLRRRELLEKLLXKLHTA	0.996

HADDOCK[45]. Moreover, Hpepdock ranks as the top performer in a large-scale global docking study, achieving the highest success rates among 14 evaluated docking programs[46]. As shown in Supplementary Section S7, in the blind docking setting, Hpepdock scores showed only moderate ability to distinguish native interacting peptides from decoy peptide-protein pairs. This incomplete separation between positive and negative pairs implies that Hpepdock can still assign relatively favorable scores to some non-interacting decoys, leading to potential false-positive predictions.

We observed that when the binding site is known, however, Hpepdock can reliably generate bound structures for experimentally validated peptide binders, whereas PepInter provides more discriminative power even when the binding site is unknown. This indicates that docking-based methods and PepInter serve complementary roles: Hpepdock is useful for pose-level structural modeling, particularly when reliable receptor structures and binding-site information are available, while PepInter operates directly on protein and peptide sequences and is therefore suitable for large-scale protein-peptide interaction screening without requiring predefined binding sites or structural inputs. In practical workflows, PepInter can be used as a front-end sequence-based screening tool to prioritize candidate peptides, followed by structure-based docking for binding-pose refinement and structural interpretation.

Discussion

Protein-peptide interactions (PpIs) play crucial roles in immune regulation, signal transduction, and peptide drug design, underscoring the need for efficient and accurate prediction methods for cutting-edge biomedical research. However, unlike protein-protein interactions where binding partners are relatively rigid, PpIs exhibit greater conformational dynamics, shallow binding interfaces, and frequent structural ambiguity. These characteristics result in substantially fewer experimentally determined structures and affinities, presenting significant challenges for data-driven modeling of PpIs[47,48]. In recent years, extensive efforts have been devoted to developing PpI prediction methods. These approaches primarily represent the protein and peptide sequences independently, employing various fusion strategies to try to capture their associations and finally relate them to downstream classification and affinity regression tasks. Notably, Transformer- and BERT-based large language models have demonstrated success by tokenizing sequences and modeling inter-sequence relationships, while recent ESM2 and related pretrained modules have further enhanced prediction accuracy through learning structured features directly from protein sequences[49,50]. Despite the promising performance achieves by existing methods, there remains substantial limitations such as time-consuming and high cost that prevent them from fully meeting real-world application demands. Interactions within the protein-peptide complexes are key to peptide biological function, suggesting that learning authentic interaction patterns is critical for identifying true binding determinants.

In this study, we propose PepInter, a PpI representation learning framework designed to capture authentic interaction patterns from protein-peptide pairs. To address the problem of limited experimental data, we generated a large dataset of pseudo protein-peptide pairs, ProtFragDB, subsequently filtered using Rosetta energy constraints for computational validation. Totally, 9 million high-quality pseudo peptide-protein fragment pairs were obtained, resulting in a training set of 8,983,271 pairs for training and 22,117 pairs for testing in the pretraining phase. PepInter was subsequently fine-tuned on two downstream prediction tasks: PpI classification and protein-peptide binding affinity prediction. For the PpI classification task evaluated on PepInterDB, PepInter consistently outperforms traditional machine learning baselines and state-of-the-art deep learning methods. Specifically, it achieves an ACC of 0.85, REC of 0.75, F1-score of 0.84, AUC of 0.94, and AUPR of 0.95, yielding substantial improvements over the strongest method. For the binding affinity regression task evaluated on PepAffDB, PepInter achieved the lowest prediction errors among all competing methods, with an RMSE of 1.02, MAE of 0.77, and the highest coefficient of determination ($R^2 = 0.65$), outperforming established affinity prediction models. Two real-world case studies further confirm the reliability and practical utility of PepInter. In the semaglutide study, PepInter was retrained on a filtered dataset excluding semaglutide-like peptides and GLP-1R-like proteins, and subsequently applied to screen 2940 candidate proteins. PepInter successfully ranked GLP-1R within the top 5% of candidates for all seven semaglutide analogs, with predicted interaction probabilities at or above 0.98 and an overall AUC of 0.98, demonstrating its strong predictive capability for peptide drug target identification. In the parathyroid hormone (PTH) case study, PepInter consistently identified PTH1R as a top-ranked target for multiple PTH and PTHrP analogs, achieving an AUC of 0.97, further validating its robustness and generalizability.

PepInter achieves significant performance gains across multiple downstream tasks, validating the effectiveness of leveraging pseudo protein-peptide data derived from structural sources together with a cross-chain representation learning strategy. The diverse peptide fragments generated under Rosetta energy constraints enable the model to learn more comprehensive interfacial energy features, thereby improving its ability to characterize key binding determinants. During pre-training, PepInter captures cross-chain contextual dependencies between proteins and peptides, which facilitates more accurate identification of potential binding regions and critical peptide motifs. This design also explains why the model preserves sensitivity to structural interaction patterns even when only single-chain sequence information is provided at inference time. Moreover, the robust performance observed under highly imbalanced and realistic evaluation scenarios indicates that PepInter goes beyond learning superficial sequence co-occurrence patterns, instead inferring functionally relevant residues and interfacial energy signals that govern PpIs.

To evaluate how PepInter performs on peptides with different disorder propensities, we stratified the test peptides into ordered and disordered subsets based on AIUPred-predicted disorder scores and compared the model's performance across these two groups (in ACC and F1, ordered: 0.873/0.833; disordered: 0.851/0.836). The results show that PepInter performs consistently across both groups, suggesting that it effectively leverages large-scale pseudo protein-peptide pairs derived from protein-protein interfaces. Since intrinsically disordered peptides constitute a large proportion of signaling peptides and molecular recognition motifs in vivo, the comparable performance of PepInter on both ordered and disordered subsets suggests its potential applicability to biologically relevant peptide discovery. This behavior has both advantages and limitations. On one hand, PepInter's high accuracy appears robust to the peptide's disorder propensity. On the other hand, pseudo pairs represent peptides only in their bound-like conformations and do not explicitly encode the entropic cost of folding from a disordered state, so PepInter is more likely to learn sequence-interface compatibility patterns than the full biophysical complexity of peptide folding and recognition, particularly for highly flexible or intrinsically disordered peptides. Such a limitation, which is inherent to the pseudo peptide-protein pairs, can likely only be mitigated by supplementing the training data with richer conformational information, for example, experimentally resolved peptide-protein complexes or physical simulation data that better capture folding and disorder.

Compared with traditional methods that rely solely on sequence similarity or local convolutional strategies, our approach retains superior performance when generalizing to unseen proteins or peptides,

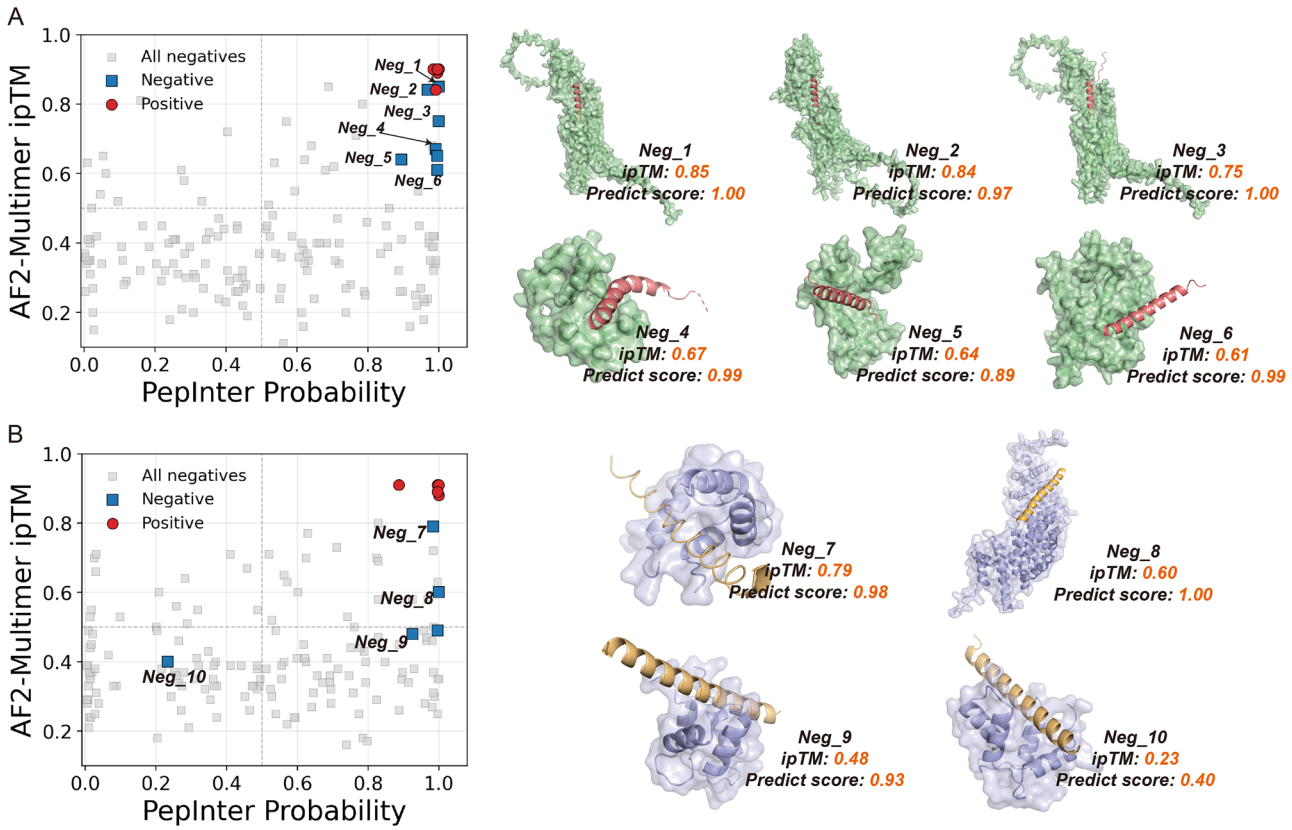


Fig. 6 | Comparison of PepInter prediction probability and AlphaFold-Multimer ipTM scores in focused case-level structural validation. A Semaglutide-GLP-1R case study. Each data point represents a peptide–receptor pair. Red circles indicate known positive pairs between GLP-1R and seven GLP-1R-binding peptides. Blue squares indicate representative highlighted hard-negative pairs selected from non-cognate receptors, and gray squares indicate all negative pairs generated from the selected non-cognate receptors to illustrate the overall distribution. **B** PTH/PTHrP–PTH1R case study. Red circles indicate known positive pairs involving cognate PTH1R, blue squares indicate representative highlighted hard-negative pairs selected from non-cognate receptors, and gray squares indicate all negative pairs. In both panels, the vertical dashed line denotes PepInter probability is 0.5, and the horizontal dashed line denotes AlphaFold-Multimer ipTM is 0.5. Hard negatives were selected from high-scoring non-cognate receptors predicted by PepInter, thereby representing challenging cases for sequence-based discrimination. The comparison illustrates that AlphaFold-Multimer ipTM provides complementary structural confidence for evaluating PepInter-prioritized candidates, rather than serving as a full-scale benchmark. For illustrative purpose, the Neg_10 example, which receives low PepInter and low ipTM scores, is shown.

demonstrating stronger cross-domain transferability-an essential property for peptide drug virtual screening and lead compound optimization. Nevertheless, this study still has some limitations: (i) the current model relies heavily on pseudo energy labels in the pre-training data, and its generalizability to real experimental affinity requires further assessment; (ii) although the model can identify important residues, the underlying binding mechanisms are not yet clearly interpretable; (iii) the sliding-window strategy employed during data construction may overlook long-range binding modes arising from conformational changes; (iv) explicit structural modeling was not incorporated, suggesting that future work may benefit from integrating AlphaFold-based structure prediction and 3D graph neural networks to enhance geometric awareness; and (v) the current framework focuses on linear peptides composed of standard amino acids, and therefore does not explicitly model cyclic peptides or chemically modified peptides, which represent an important class of therapeutic peptides with distinct structural constraints and physicochemical properties.

In the future, we will focus on the following directions: (i) integrating 3D structural constraints and dynamic simulation data to better capture peptide binding mechanisms; (ii) introducing automated hyperparameter optimization and continual pre-training strategies to enhance the model's adaptability to new structural domains; (iii) extending the framework to support cyclic peptides, non-canonical residues, and chemically modified peptides; and (iv) applying the framework to peptide drug virtual screening and binding affinity optimization, complemented by wet-lab experiments to validate biological efficacy. These directions aim to improve the modeling of flexible peptides and account for entropic effects, thereby further enhancing the accuracy and generalizability of protein-peptide interaction predictions while providing practical pathways for computational peptide drug design.

Methods

Pseudo protein-peptide pretraining dataset - ProtFragDB

ProtFragDB was constructed using protein complexes from the PPI3D database[51] which contains more than 891,000 experimentally determined PDB Protein-Protein Interaction (PPI) pairs. Samples were selected to ensure high-quality, nonredundant data, using the following criteria: (i) structural resolution better than 4.0 Å; (ii) protein sequence length fewer than 1000 amino acids; and (iii) a maximum sequence identity of 40%, as determined with MMseqs2 (version Release 18-8cc5c)[52], to reduce redundancy among PPI pairs. After filtering and clustering, a total of 28,916 nonredundant PPI pairs were obtained. Subsequently, the Rosetta Peptiderive protocol (Rosetta 3.14) was applied to these PPIs to generate pseudo protein-peptide fragment pairs. For each protein-protein complex, one chain was fixed as the receptor protein, while the interacting partner chain was systematically decomposed into linear peptide fragments using a sliding-window strategy with fragment lengths ranging from 5 to 50 amino acids[53]. For each generated pseudo peptide-protein pair, Peptiderive computed an interface energy score to quantify the energetic contribution of the peptide fragment to the interaction interface. The pairs associated with abnormal or erroneous interface scores were discarded. To further eliminate redundancy, the resulting pseudo peptide-protein pairs were clustered again

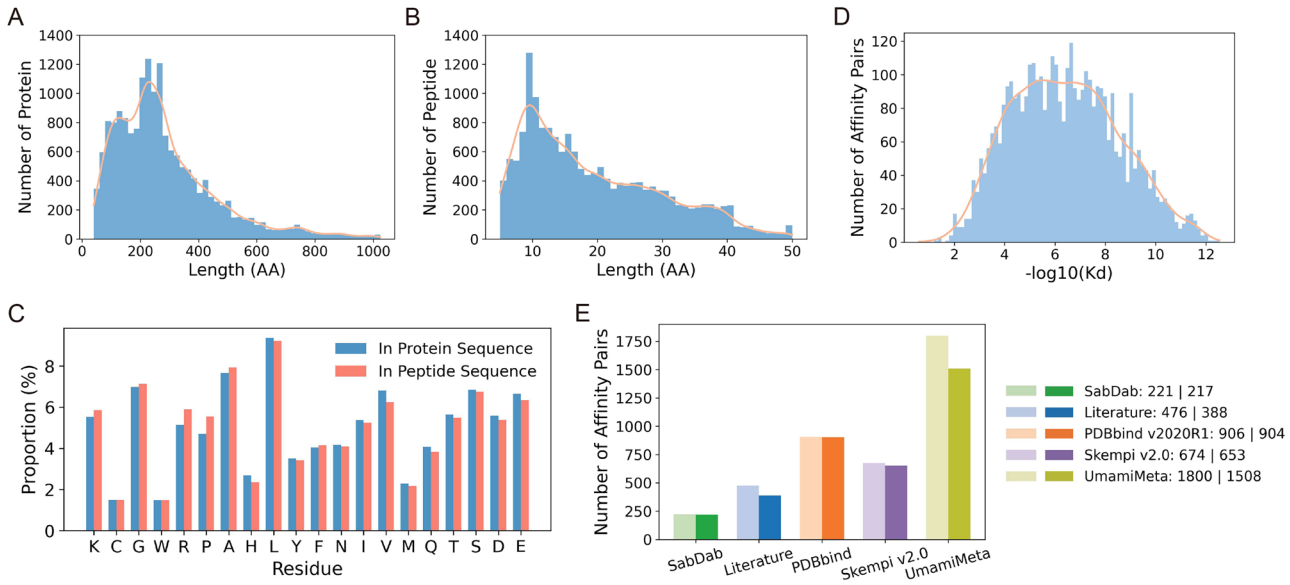


Fig. 7 | Characteristics of PepInterDB and PepAffDB. A Length distribution of protein sequences in PepInterDB. **B** Length distribution of peptide sequences in PepInterDB. **C** Comparison of amino acid composition between protein and peptide sequences in PepInterDB. **D** Distribution of binding affinities, expressed as $-\log_{10}(K_d)$, in the final PepAffDB comprising 3670 affinity pairs. **E** Number of protein-peptide affinity records in PepAffDB, before and after excluding peptides shorter than 5 residues across different data sources.

using MMseqs2 with a 40% sequence identity threshold. Sequences belonging to the same cluster were strictly assigned to the same data split to prevent information leakage between the training and validation sets. After deduplication, approximately 9 million high-quality pseudo peptide-protein fragment pairs were retained, including 8,983,271 pairs for training and 22,117 pairs for validation.

Experimental protein-peptide interaction dataset - PepInterDB

The protein-peptide interaction dataset namely PepInterDB was constructed by systematically scanning all protein complex structures deposited in the RCSB Protein Data Bank (PDB, accessed in September 2025)[54]. A Python-based (Python version 3.12) pipeline was developed to identify interacting chain pairs, where two chains were considered to form an interaction if any pair of heavy atoms between them was within a distance threshold of 5 Å. To specifically capture protein-peptide interactions, interacting chain pairs were further filtered by chain length, requiring the protein chain to contain 40-1000 residues and the peptide chain to contain 5–40 residues. Redundant interaction pairs with identical protein-peptide sequence combinations were first removed. In addition, sequence redundancy was further reduced using MMseqs2 clustering at a high sequence identity threshold (80%), to eliminate highly similar protein or peptide sequences while preserving sufficient data diversity. Following these filtering steps, a total of 8527 non-redundant protein-peptide interaction pairs were obtained. The dataset was randomly split into a training set containing 6630 samples and a validation set containing 1,897 samples. An equal number of negative samples were additionally generated by first removing all known interaction pairs and then randomly sampling non-interacting pairs from the remaining candidates, yielding a final dataset of 17,054 protein-peptide pairs. The protein and peptide length distributions of PepInterDB are shown in Fig. 7A, B, respectively. Figure 7C compares the amino acid composition of protein and peptide sequences, revealing broadly similar distributions with subtle differences across residues. In these datasets, sequences containing non-standard amino acids or ambiguous amino acids (e.g., X, B, Z, and J) were retained provided that the number of such residues did not exceed half of the total sequence length; otherwise, the corresponding samples were removed. This strategy maximizes data utilization while maintaining overall sequence quality. Notably, non-standard amino acids were allowed only in the training set, whereas the validation set contained exclusively standard amino acid residues. This design intentionally introduces mild noise during training, which is expected to improve model robustness and generalization performance.

Protein-peptide affinity dataset - PepAffDB

Several works have previously reported protein-peptide affinity dataset, but the existing datasets are often insufficient in terms of quality and scale. To better evaluate the generalization ability of the proposed model, a protein-peptide affinity dataset, PepAffDB, was constructed by collecting data from multiple sources, including PDBbind v2020R1[54], SKEMPI v2.0[55], SabDab[56], UmamiMeta[57] and published literature. Specifically, a total of 937, 246, 2069 and 1869 protein-peptide affinity records were collected from PDBbind v2020R1, SabDab, UmamiMeta and SKEMPI v2.0, respectively, after applying dataset-specific filtering criteria. In addition, 481 protein-peptide records were manually curated from more than 500 published articles through a systematic literature search using keywords related to "protein-peptide affinity". These records include experimentally synthesized peptides, sequence variants and other peptide modifications, most of them do not have resolved structural information. After integrating all sources, extensive data cleaning, filtering, and deduplication were performed, resulting in 4077 affinity records. To ensure data reliability, peptides shorter than 5 residues were excluded, yielding a final dataset of 3670 high-quality protein-peptide pairs with experimental binding affinity values, referred to as *affinity pairs* for simplicity. Figure 7D presents the distribution of binding affinities in the final PepAffDB dataset, and Fig. 7E summarizes the number of protein-peptide affinity pairs. Moreover, PepAffDB were clustered using MMseqs2 with an 80% sequence identity threshold. Sequences belonging to the same cluster were strictly assigned to the same data split to prevent information leakage, resulting in 2698 pairs for training, 671 pairs for validation, and 301 pairs for testing. Affinity measurements reported in different units, including K_d, K_i, and IC_{50}, were retained as provided by the original sources. The affinity values were converted into a unified logarithmic scale to facilitate model training and evaluation, while records lacking sufficient information for reliable conversion were excluded. The final dataset covers a broad range of binding strengths spanning multiple orders of magnitude, enabling robust assessment of model performance across diverse affinity regimes. The dataset is available on GitHub.

Cluster-based data splitting strategy

Because a peptide can bind to multiple similar proteins and a protein can also interact with multiple similar peptides, this pattern often contains substantial redundancy. To eliminate the potential bias introduced by such redundancy and to ensure a fair comparison, we followed the same strategy as in CAMP[42] and MONN[58], which employed a cluster-based cross-validation settings to evaluate model performance. Specifically, we computed pairwise similarity scores using the Smith-Waterman local alignment algorithm (https://github.com/mengyao/Complete-Striped-Smith-Waterman-Library). Based on these scores, proteins and peptides were clustered under a series of similarity thresholds (0.3, 0.4, 0.5, and 0.6), and three evaluation settings were constructed, namely "Novel Protein", "Novel Peptide", and "Novel Pair". In the "Novel Protein" and "Novel Peptide" settings, proteins and peptides were clustered based on their sequences, respectively. In the "Novel Pair" setting, each protein-peptide interaction pair was assigned to a pair cluster defined by the combination of its protein-cluster and peptide-cluster assignments. In all three settings, the data split was performed at the cluster level rather than at the sample level, ensuring that proteins, peptides, or interaction pairs belonging to the same cluster were not shared between the training and test sets.

Model pre-training

The quality of protein representations has a direct impact on the performance of downstream prediction tasks. Recently, PLMs trained in an unsupervised manner on massive sequence datasets, such as the ESM series, have become standard tools for deriving high-quality protein embeddings. ESM-C[29] has demonstrated remarkable advantages in representation learning that it outperforms ESM-2 while offering faster inference speed and lower computational cost with a comparable number of parameters. With the scale of 600 M parameters, ESM-C achieves remarkable computational efficiency comparable to that of the 300 M-parameter ESM-2 model. Given this advantage, ESM-C is adopted as the base architecture during pre-training, with a masked language modeling (MLM) strategy employed as the self-supervised learning objective on both protein and peptide sequences. The learning problem can be formulated as follows. Given a pair of protein and peptide sequences:

$$\begin{aligned} X_{prot} &= \{x_1^{prot}, x_2^{prot}, \ldots, x_{L_{prot}}^{prot}\} \\ X_{pep} &= \{x_1^{pep}, x_2^{pep}, \ldots, x_{L_{pep}}^{pep}\} \end{aligned} \tag{1}$$

where L_{prot} denotes the length of the protein sequence and L_{pep} denotes the length of the peptide sequence. We define a preprocessing template for the tokenizer to standardize the format of the input sequences, thereby facilitating clear distinction between the protein and peptide sequences within each pair. Specifically, the protein-peptide pair template is as follows:

$$\langle cls\rangle : 0\ X_{prot} : 0 \,|: 0\ \ X_{pep} : 1\ \ \langle eos\rangle :\ \ 1 \tag{2}$$

where 0 and 1 are the token-type IDs for distinguish the protein and peptide sequences, respectively. The symbol '|' is a delimiter connecting the two sequences, while <cls> and <eos> denote the start and end markers, respectively. All special symbols in the template (<cls>, <eos>, '|') are registered in the vocabulary as special tokens and are converted into their corresponding token IDs. These start and end markers enable the model to clearly recognize sequence boundaries while the delimiter marks their separation. This approach specifies the input representation of a protein-peptide pair for the model, and enforcing a consistent input format is expected to improve the consistency and stability of future downstream prediction tasks. During training, approximately 15% of amino acid residues are randomly selected as masked positions, and the model is trained to predict the true amino acid at each masked site, learning the semantic and structural patterns within the sequences. The training loss is defined as follows:

$$L_{MLM} = -\frac{1}{|M|}\sum_{i\in M}\sum_{k=1}^{V} y_{i,k} log\left(\widehat{y_{i,k}}\right) \tag{3}$$

where $|M|$ denotes the number of masked tokens; V means the size of the vocabulary; $y_{i,k}$ is an indicator function that equals 1 if the true class of token i is k, and 0 otherwise; $\widehat{y_{i,k}}$ is the probability predicted by the model that token i belongs to class k. The model can automatically capture amino acid co-occurrence preferences, local and long-range dependency patterns, as well as structural and functional information implicit in protein-peptide interactions via a self-supervised objective, thereby providing more generalizable sequence representations for downstream prediction tasks.

Fine-tuning

During the fine-tuning stage, the model is initialized with the parameter weights learned during pretraining, fully leveraging the general representation capabilities acquired from large-scale protein-peptide sequence pairs. The model is then trained in a supervised manner on the protein-peptide dataset. The inputs are the protein sequence L_{prot} and the peptide sequence L_{pep}. Depending on the task setting, the prediction $\widehat{y}$ represents either a binary classification probability or a continuous binding affinity value. For the classification task, the Focal loss[59] is adopted which is a modified cross-entropy loss function tackles extreme class imbalance, the original loss function is shown as follows:

$$\begin{aligned} L_{fl}(p_t) &= -(1-p_t)^{\gamma} log(p_t) \\ p_t &= \begin{cases} p & if\ y=1 \\ 1-p & otherwise \end{cases} \end{aligned} \tag{4}$$

where γ is set to 2. In our experiment, the α-balanced variant of the focal loss is employed:

$$L_{FL}(p_t) = -\alpha_t(1-p_t)^{\gamma} log(p_t) \tag{5}$$

where α_t is set to 0.25.

For the binding affinity prediction task, a Mixture Density Network (MDN)[60] loss is adopted to model the potentially multi-modal distribution of binding affinities. Specifically, the MDN head outputs the parameters of a K-component Gaussian mixture, including the mixture weights $\pi \in R^K$, the means $\mu \in R^{K\times D}$, and the standard deviations $\sigma \in R^{K\times D}$. The training objective is the negative log-likelihood of the weighted sum of the component distributions as follows:

$$L_{MDN}(y) = -\log\left(\sum_{k=1}^{K}\pi_k \mathcal{N}(y|\mu_k,\sigma_k)\right) \tag{6}$$

To improve robustness against label noise and hard outliers, we further apply a small-loss selection strategy during training. For each mini-batch, we first compute the per-sample MDN loss $\{l_i\}_{i=1}^{B}$, and determine a quantile threshold q_r at a predefined retention ratio r (e.g., r = 0.8). Only samples with $l_i \le q_r$ are assigned non-zero weights and used to update the model, with the weights normalized to keep the average weight close to 1. The final training objective is:

$$\mathcal{L} = \frac{1}{B}\sum\nolimits_{i=1}^{B} w_i L_{\mathrm{MDN}}(y_i) \tag{7}$$

where $w_i \in \{0, 1\}$ indicates whether the *i-th* sample is selected. This small-loss trick adaptively filters out high-loss samples which are potentially noisy, stabilizing training and yielding more robust estimates of the protein-peptide behavior.

Prediction head of protein-peptide interaction classification and affinity regression

The protein-peptide interaction prediction task aims to determine whether a given protein-peptide pair interacts, by mapping the input sequence pair to a binary label in {0, 1}. In this task, the classification head is implemented as a two-layer feed-forward network. Given a feature representation $F \in R^{d_m}$, the hidden representation is computed as:

$$Z = \mathrm{LayerNorm}\left(\mathrm{GELU}\left(w_1 F + b_1\right)\right) \tag{8}$$

where the Gaussian Error Linear Unit (GELU) serves as the activation function, providing a smooth, non-linear transformation, whereas w_1 and b_1 denote the weight matrix and bias vector of the first fully connected layer, respectively. The final output logits are then obtained by:

$$\begin{aligned} Pred &= w_2 Z + b_2 \\ \widehat{y} &= \begin{cases} 1, & pred > 0.5, \\ 0, & otherwise \end{cases} \end{aligned} \tag{9}$$

where w_2 and b_2 are the weight matrix and bias vector of the output layer. For binary interaction prediction, a sigmoid function is applied to produce the interaction probability $\widehat{y}$.

For the binding affinity prediction task, we employ a MDN head to model the potential binding affinities. Given a feature representation $F \in R^{d_m}$, the final training loss is computed as a normalized weighted average over the selected samples. At inference time, the predicted binding affinity $\widehat{y}$ is obtained as the expectation of the mixture distribution:

$$\widehat{y} = \sum_{k=1}^{K} \pi_k \mu_k \tag{10}$$

where K denotes the number of Gaussian components, π_k represents the mixture weight of the *k-th* component, and μ_k denotes the corresponding component mean.

Performance metrics

This study employs widely used binary classification metrics to comprehensively evaluate the performance of various machine learning models in PpIs classification tasks. These metrics include accuracy (ACC), recall (REC), F1-score, the area under the ROC curve (AUC), and the area under the precision-recall curve (AUPR). Overall, these metrics not only assess the overall predictive performance but also evaluate their ability to maintain balance between positive and negative classes. The definitions of these metrics are as follows:

$$ACC = \frac{TP + TN}{TP + FP + TN + FN} \tag{11}$$

$$PRE = \frac{TP}{TP + FP} \tag{12}$$

$$REC = \frac{TP}{TP + FN} \tag{13}$$

$$F1 = 2 \times \frac{PRE \times REC}{PRE + REC} \tag{14}$$

In this context, True Positive (TP) and True Negative (TN) represent the number of positive samples correctly identified by the model, the number of negative samples that are correctly classified, respectively. False Positive (FP) and False Negative (FN) refer to negative samples that are incorrectly predicted as positive, positive samples that are incorrectly classified as negative, respectively.

Similarly, the root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2) are used in the protein-peptide affinity regression task. RMSE and MAE reflect the absolute magnitude of prediction errors, with smaller values indicating better performance. R^2 measures the explanatory power of the model, and values closer to 1 indicate better fitting capability. Their definitions are as follows:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2} \tag{15}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \widehat{y}_i\right| \tag{16}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{N}\left(y_i - \bar{y}\right)^2} \tag{17}$$

where y_i and $\widehat{y}_i$ denote the true and predicted values, respectively; $\bar{y}$ represents the mean of the true values, and N is the number of samples.

Pre-training configurations and learning hyperparameters

In the construction of ProtFragDB, the Rosetta Peptiderive protocol was employed to identify protein-protein interaction interfaces from given protein structures and to predict peptide fragments with potential interface-binding capability. In Peptiderive, the primary parameters "*restrict_receptors_to_chains*" and "*restrict_partners_to_chains*" were specified to define the receptor protein chains and the corresponding partner chains from which peptides were derived. The option "*dump_peptide_pose*" was set to true to output the conformations of the derived peptide fragments, while "*dump_prepared_pose*" was enabled to retain the preprocessed protein-peptide complex structures prior to docking. All remaining parameters were kept at their default values.

During the pre-training phase, 36 Transformer layers were employed, with each layer containing 18 attention heads. The model was trained for three steps with a batch size of 128. The maximum sequence length was set to 1000 for proteins and 50 for peptides. The hidden dimension was 1152 and AdamW[61] was used as the optimizer with an initial learning rate of 1×10^{-4}. The maximum number of training steps was 10,000. Pre-training was conducted on three NVIDIA A800 80GB PCIe GPUs and converged in approximately 72 h. The pretraining loss is shown in Supplementary Section S2.

Fine-tuning configurations for protein-peptide interaction and affinity predictions

The protein-peptide interaction prediction model was developed by fine-tuning PepInter on the experimental PepInterDB dataset for protein-peptide interaction prediction. Protein and peptide sequences were encoded separately using the pre-trained ESM-C encoder. The resulting residue-level contextual representations were subsequently fused and fed into a binary classification prediction head to output the binding probability. Full-parameter fine-tuning was used to effectively transfer the pre-trained representations to the downstream task. The batch size was set to 24, the maximum number of prediction epochs was 5, and the learning rate was set to 1×10^{-4}.

Similarly, for the protein-peptide affinity prediction task, we fine-tuned the model using the experimental protein-peptide affinity dataset PepAffDB. The fused representations were fed into a regression prediction head to output quantitative binding affinity values. The fine-tuning configuration was consistent with that used in the interaction prediction stage, with a maximum of 50 prediction epochs and a learning rate of 1×10^{-4}.

Data availability

The datasets used in this study are publicly available. The PPI3D is available at https://bioinformatics.lt/ppi3d/start[51], and PepInterDB is available

through RCSB PDB (accessed in September 2025) at https://www.rcsb.org/[54], and PepAffDB is available through PDBbind v2020R1 at https://www.pdbbind-plus.org.cn/[54], SKEMPI v2.0 at https://life.bsc.es/pid/skempi2[55], SabDab at https://sabdab.opig.stats.ox.ac.uk/[56], UmamiMeta at https://hwwlab.com/Webserver/umamimeta[57]. The PDB file of 3EJH, 9DPF is available from the PDB dataset https://www.rcsb.org/. The data in this study have been publicly deposited to Zenodo at https://doi.org/10.5281/zenodo.21715291[62]. The source data for Figs. 3–7 are provided in the corresponding Supplementary Data (Fig. 3–Fig. 7 **sheets**).

Code availability

The code for PepInter is publicly available via GitHub at https://github.com/zhanxinke/PepInter.git and has been archived on Zenodo at https://doi.org/10.5281/zenodo.21717699[63].

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Author contributions

X.Z. and S.Z. collected the data and wrote the paper, T.B. and Z.B. wrote the program; T.L. and S.L. plotted the figures and cleaned the data. Shirley W.S. revised the paper.

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Competing interests

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Additional information

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